MIL-C-51542(EA)
29 September 1988
SUPERSEDING
(See section 6)

MILITARY SPECIFICATION

CONVERTER, FREQUENCY, STATIC, XM5 (FOR M56, XM59, OR XM62 FILTER UNITS)

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a static frequency converter from a 208 volt (V), 3-phase, 60 Hertz (Hz) power source.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards and handbooks. The following standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-ST'D-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production.

: Beneficial comments (recommendations, additions, deletions) and any Pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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HANDBOOKS

MILITARY

MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-19-6425 - Converter, Frequency, Static.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-6425 - Converter, Frequency, Static.

(Copies of standards, handbooks, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence, Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-6425 and on the subsidiary drawings shall conform to the specifications listed thereon, and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the converter shall conform to the specifications and drawings listed on Drawing 5-19-6425 and subsidiary drawings.

3.2 Manufacture and assembly. The converter shall be manufactured and assembled in accordance with Drawing 5-19-6425.

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3.3 Performance.

3.3.1 Cycling burn-in. The converter shall be subjected to a cycling burn-in which shall consist of 3 hours continuous operation under no load conditions and 3 hours on-off operation in accordance with 4.4.4.1.

3.3.2 Operational characteristics. The converter shall conform to the schematic representations specified in Drawings 5-19-6492 and 5-19-6437 and shall perform as specified in 3.3.2.1 through 3.3.2.6 when the applicable voltages, voltage phases, and load are applied to the specific junctions on the static frequency converter.

3.3.2.1 DC voltage. The test point DC voltages from test points 1, 2, and 3 to test point 4 shall be between 10.75 and 13.25 when tested as specified in 4.4.4.2(b).

3.3.2.2 AC voltage. The test point AC voltage between test points 5 and 6 shall be greater than 2 volts when tested in accordance with 4.4.4.2(C).

3.3.2.3 Phasing. The phasing of the output shall be phase 1 on pin A, phase 2 on pin B and phase 3 on pin C of connector J8, when tested in accordance with 4.4.4.2(d).

3.3.2.4 Output voltage (line to line). The output line to line voltage with load applied (see 6.3) shall be 208 + 15 Volts, root-mean-square (VRMS), when tested in accordance with 4.4.4.2(e).

3.3.2.5 Output voltage (line to neutral). The output line to neutral voltage with load applied (see 6.3) shall be 120 ± 10 VRMS when tested in accordance with 4.4.4.2(f).

3.3.2.6 Output frequency. The output frequency of each phase shall be 400 ± 10 Hz when tested in accordance with 4.4.4.2(g).

3.4. Reliability.

3.4.1 Preproduction reliability qualification. Prior to the start of regular production, the preproduction sample converters shall demonstrate acceptable reliability when tested in the configuration specified in 4.4.4.1 in accordance with task 302 of MIL-STD-781, using test plan IV of MIL-HDBK-781 with minimum test mean time between failures (01) equal to 934 hours. An accountable failure shall be defined as any deviation from the requirements specified in 3.3.2.1 through 3.3.2.6.

3.4.2 Production reliability. The production reliability acceptance sample for each lot shall demonstrate acceptable reliability when tested in the configuration specified in 4.4.4.1 in accordance with task 302 of MIL-STD-781, using test plan IV of MIL-HDBK-781 with a minimum test mean-time-between-failures (01), equal to 934 hours. An accountable failure shall be defined as any deviation from the requirements specified in 3.3.2.1 through 3.3.2,6,

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3.5 Workmanship. The converter shall be free from chipped or bent metal sections, and foreign matter (dirt, oil, or viscous materials).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. Packaged sample converters shall be examined in accordance with the classification of defects (4.4.3.4(b)). After completion of the packaging inspection, the sample converters shall then be unpackaged and examined in accordance with the classification of defects (4.4.3.4(a)).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction packaged sample of four converters shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection Procedures.

4.3.2.1 For examination. The packaged preproduction inspection sample of converters shall be inspected for all the requirements of the drawings and this specification.

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4.3.2.2 For test. The unpackaged preproduction sample of converters shall be tested in accordance with 4.4.4.1 and 4.4.4.2 for conformance to 3.3. Then the converters shall be tested in the configuration specified in 4.4.4.1 for conformance to 3.4.1.

4.3.3 Acceptance/rejection criteria: Each converter in the packaged preproduction sample shall meet the examination specified in 4.4.3.4. When tested in accordance with 4.3.2.2 the unpackaged preproduction inspection sample shall meet the requirements of 3.4.1 and the criteria of Test Plan IV of MIL-HDBK-781 to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the converters of one type produced by one manufacturer, from the same materials, at the same locations, under essentially the same manufacturing conditions, and at essentially the same time.

4.4.2 Sampling.

4.4.2.1 For examination. The unpackaged sample converter for examination shall be conducted in accordance with MIL-STD-105. For the repair part acquisition, the packaged converter for examination shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Each converter shall be tested.

4.4.2.3 For reliability production acceptance. A sample of four converters shall be taken randomly from each production lot for reliability production acceptance testing.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample converters shall be examined in accordance with 4.4.3.4(a) of the classification of defects. For the repair part acquisition only, the packaging inspection (4.4.3.4(b)) shall also be Conducted.

4.4.3.2 For test. Each converter shall be tested for cycling burn-in and performance prior to packaging. Any converter failing any of the performance requirements shall be rejected.

4.4.3.3 For reliability production acceptance test. The reliability production acceptance test sample of unpackaged converters shall be tested in accordance with 4.4.4.1 and 4.4.4.2 for conformance to 3.3. Then the converters shall be tested in the configuration specified in 4.4.4.1 for conformance to 3.4.2.

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4.4.3.4 Classification of defects.

(a) Converter, frequency, static (Dwg 5-19-6425).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing, incorrect or incorrectly assembled
102	Damaged wires
103	Improper soldering
104	Workmanship (3.5)

(b) Packaging inspection (SPI P5-19-6425) (see section 5).

Categories	Defects
Critical:	None defined
Major	AQL 2.5 percent defective
101	Item not completely cleaned/dried
102	Cushioning pads missing or not as specified
103	Wrap missing or not as specified
104	Unit or shipping container marking incorrect, missing or illegible
105	Unit or shipping container and closure not as specified
106	Unit or shipping container damaged

4.4.4 Tests. Cycling burn-in and performance tests shall be conducted as follows:

4.4.4.1 Cycling burn-in test.

(a) Position the converter on a work bench so that the electrical connectors are readily accessible. Open the protective cover on the circuit breaker and fault isolation test points. Open the circuit breaker. Connect the converter to a 208 V, 60 Hz, 3-phase power source through a three pole relay as follows:

Phase 1 to pin A of J7
Phase 2 to pin B of J7
Phase 3 to pin C of J7

CAUTION: Pin H of J7 must be grounded.

(b) Connect the coil of the relay to a timer so that it may be energized at the rate of once every minute.

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(c) Close the circuit breaker on the converter and energize the relay. After approximately 5 seconds, the converter will start as indicated by a high-frequency buzzing sound. Operate the converter in this configuration (no load) for 3 hours. Then actuate the timer so that the converter cycles, on and off, and operate the converter in this configuration for 3 hours. For reliability testing, the cycling shall continue until the reliability is demonstrated.

4.4.4.2 Performance. After the cycling burn-in test, the converter shall be inspected to assure that it is still cycling properly and that the circuit breaker is set. Then:

(a) With the power applied as specified in 4.4.4. l(a) operate the converter and make the following measurements using standard voltmeters and frequency counters. The voltmeter used in 4.4.4.2(e) and 4.4.4.2(f) must be a true root-mean-square reading meter. An average responding meter with a root-mean-square compensated scale is not acceptable because of the pseudo square wave output of the converter.

(b) DC voltage. Measure in turn the DC voltage from test points 1, 2, and 3 to test point 4. This voltage shall be as specified. This test is run with no load on the converter.

(c) AC voltage. Measure the AC voltage between test points 5 and 6. This voltage shall be as specified. This test is run with no load on the converter.

(d) Phasing. Connect a 7 to 8 amp per phase load (see 6.3) to the output of the converter as follows:

Phase 1 to pin A of J8
Phase 2 to pin B of J8
phase 3 to pin C of J8
Neutral to pin D of J8

CAUTION: The output neutral of the converter must be isolated from ground (see 6.3).

Test the output phasing of the converter by observing that the applied load rotates in the proper direction when possible, or by displaying the phases in pairs on a dual trace oscilloscope for conformance with specified requirements.

(e) Output voltage. With the load applied (4.4.4.2(d)), measure the output voltage of the converter from line to line, i.e. A to B, B to C, and A to C on connector J8. This voltage shall be as specified.

(f) Output voltage. With the load applied (4.4.4.2(d)) measure the output voltage of the converter from line to neutral, i.e. A to D, B to D, and C to D on connector J8. This voltage shall be as specified.

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(g) Output frequency. With the load applied (4.4.4.2(d)) measure the output frequency of the converter on each phase. This frequency shall be as specified.

5. PACKAGING

5.1 Interplant shipment (see 6.4). The converter shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in accordance with FED. TEST METHOD STD. NO. 123, and shipping containers shall include the contractor's name, address, contract number, and date of manufacture.

5.2 Repair parts (see 6.2). When this item is being procured for storage and issue as a repair part, unit packing, packing, and marking shall be as specified in SPI P5-19-6425.

6. NOTES

6.1 Intended use. This specification covers a static converter to enable an M56, XM59 or XM62 collective protection filter unit to operate from a 208 V, 60 Hz, 3-phase power source.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether interplant shipment or repair parts packaging is required.
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Test load. It is required that the performance test specified in 4.4.4.2 be conducted with a 7 to 8 amp load on the output of the converter. It is necessary that the starting surge current of this load not exceed 35 amps for 5 seconds. It has been found that the main fan (Dwg 5-19-6137) from an XM62 filter unit, when operated at a free flow condition, satisfies the load requirements for this test. In addition, the rotational direction of the impeller can be observed to verify proper phasing of the converter output as required in 4.4.4.2(d). Neutral is isolated from ground (chassis) in this fan (4.4.4.2(d)). For safety, a separate line should be run from fan chassis to earth ground. If a load other than a 5-19-6137-20 main fan is used, it must be determined whether neutral is common with chassis in the load and caution must be exercised to maintain isolation between the converter output neutral and ground.

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6.4 Interplant shipment. Packaging and packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.5 Supersession data. This specification supersedes Chemical Research and Development Center Purchase Description EA-C-1133, dated 29 July 1980.

6.6 Subject term (key word) listing.

Converter
Filter, gas
Filter unit

Custodian:

Army - EA

Preparing activity:

Army - EA

Project 4240-A958

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-51542(EA)	CONVERTER, FREQUENCY, STATIC, XM5 (FOR M56, XM59, OR XM62 FILTER UNITS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-C-51542(A)
NOTICE 1
27 December 1995

MILITARY SPECIFICATION

CONVERTER, FREQUENCY STATIC, XM5 (FOR M56, XM59, OR XM62 FILTER UNITS)

This notice should be filed in front of MIL-C-51542(EA) dated 29 September 1988.

MIL-C-51542(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing actitity:

Army - EA

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